AI Customer Service Agent Value Evaluation

Shipment enquiries and delay handling

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We will evaluate the value created by an AI agent handling **shipment-status and delay enquiries**. Our system will connect service outcomes, quality, human work and full costs to an agreed comparison.

1 Chosen workflow

We have selected shipment enquiries for our first prototype: receive the request, identify the shipment, check operational status, respond or create an exception case, escalate when needed, then record follow-up and outcome. We will assess the whole enquiry, including human corrections and repeat contacts.

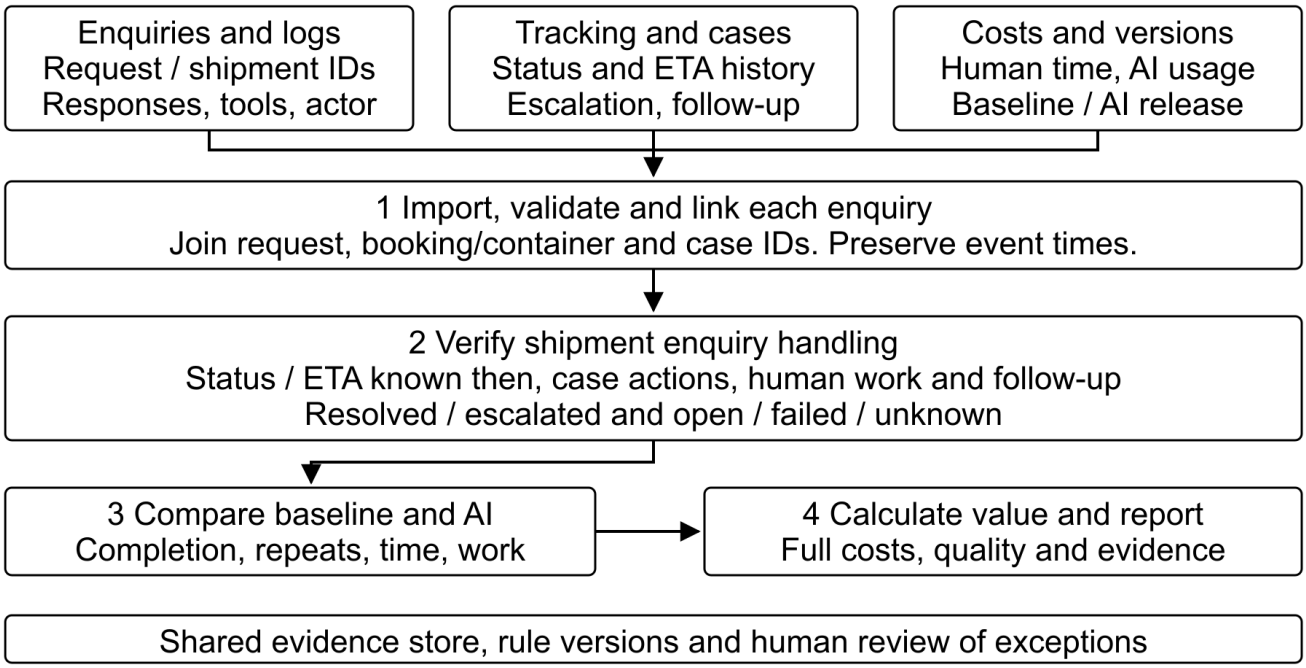
CMA CGM is our first pilot opportunity. Its public announcement describes related cargo-release and documentation enquiries. [1] We still need to confirm the accessible team's process and data. Our scope covers enquiry handling. Any effect on physical delivery requires separate evidence.

2 Research focus

We will test whether linked evidence reduces unsupported outcome and value claims when records are missing, conflicting or show human rescue. Existing AI measurement tools provide a starting point. [2] Our advantage remains to be demonstrated.

3 Architecture for shipment enquiries

Our architecture links shipment enquiries, tracking history, case records and costs. We will validate and assemble this evidence, verify enquiry outcomes, compare the selected workflows and produce an assessment of AI value.



Implementation: we propose Python, SQLite, FastAPI and Streamlit. We will start with structured test records and rules, then validate any free-text claim extraction separately.

4 Shipment data and preparation

We will request authorized CSV or JSON exports first and use simulated records until access is available. Tracking-event standards offer a mapping reference. [3] The service flow can include backend lookup, escalation and logging. [6] Exact company systems and API access remain to be confirmed.

Data	Essential fields	Potential source
Enquiry and agent	Request ID, shipment IDs, times, response, tool result, agent version	Support records and agent traces
Shipment history	Booking/container, status, ETA, event time, recorded time, snapshot ID	Tracking exports or authorized operational feed
Cases and follow-up	Linked case ID, reason, owner, handoff, updates, closure, repeat contact	Case system and interaction history
Human work	Request/case ID, activity, active duration, correction or rescue	Handling tools or measured activity
Costs and rules	AI usage/charges, labor rate, setup cost, permissions, service criteria	Finance, workflow owner and policy records

Preparation: with the workflow owner, we will agree mappings, permissions, authoritative sources and observation windows. We will validate fields and times, remove export duplicates, preserve retries and link request, shipment and case IDs into enquiry histories. We will reconcile counts and check sample histories before automating the agreed process.

Shipment checks: we will compare claims with the status/ETA available when the request was handled, check required case actions and follow-up, and record human rescue. We will distinguish resolved, escalated/open, failed and unknown enquiries. Missing historical snapshots restrict past-answer verification. A created ticket alone does not prove resolution.

5 Outputs and value comparison

We will report verified resolution, AI-only handling, incorrect or outdated information, required-action completion, handoffs, repeat contacts, human minutes, elapsed time, evidence coverage and cost per verified resolution. Open and unknown cases will remain visible.

For deployment value, we compare the existing workflow with the agent-enabled workflow. For an integrator enhancement, we compare the earlier and updated agent. We will choose one first comparison and replay equivalent test cases. Field effects require a credible production comparison. [4]

We will include all scoped attempts, technology, setup, review and correction costs, while assessing service quality. Released staff time and realized cash savings will be reported separately. [5]

6 First month prototype and validation

We will build the shipment-enquiry evaluator using file imports, linked evidence, structured outcome checks, one comparison, cost calculations, a simple review screen and exports. A simulated shipment backend and case history will supply development data. Live connectors and voice processing will follow later.

We will test normal tracking, delay escalation, stale status, wrong-shipment linkage, human rescue, repeat contact and missing records against independently checked expected results. We will compare with agent-reported success and a strong independent SQL/code analysis using the same evidence.

We will measure false resolution claims, accuracy, evidence coverage, economic error and total preparation/review effort. The demonstration will establish technical behavior. Real company ROI requires field evidence.

Week	Main deliverable
1	Define shipment-enquiry states and comparison. Map fields. Build simulated tracking and case histories.
2	Implement imports, enquiry linkage, status/case checks, human-work and cost calculations.
3	Add comparison, review interface and exports. Test against baseline analyses.
4	Run held-out and damaged-evidence cases. Document errors and demonstrate reproducible results.

Contribution and importance: we will investigate model, retrieval, tool and human effects through valid comparisons, including interactions. [7] We will keep KPIs separate initially. Any weighted score needs agreed priorities, scales and sensitivity checks, and remains separate from financial ROI. [8]

7 Feedback requested from our advisor

Please review our choice of shipment enquiries and advise us on the service boundaries, success criteria and research question. We would appreciate your feedback on the specific architecture, data-preparation plan and first comparison. We will confirm sample-data access and process details with a CMA CGM contact.

Status: we have selected the workflow and prepared this design. Company access, exact mappings and comparison remain to be confirmed. Development and experiments have not started. Competitive advantage and real financial benefit remain unproven.

Selected references

[1] CMA CGM and HappyRobot customer service deployment

[2] Workhelix FAQ on AI measurement and comparisons

[3] DCSA Track and Trace implementation guide

[4] HM Treasury Magenta Book Annex A evaluation methods

[5] FinOps Foundation FinOps for AI overview

[6] Salesforce Architects shipping enquiry runtime example

[7] NIST factorial design example on component interactions

[8] Government Analysis Function guide to criteria and weights